



ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2011

CHEMISTRY

Higher 2

Paper 1 Multiple Choice

9647/01

22 September 2011

1 hour

Additional Materials: Multiple Choice Answer Sheet
Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC/FIN number. **DO NOT** include the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Prelim

Exam Details: H2 Chem / Paper 1

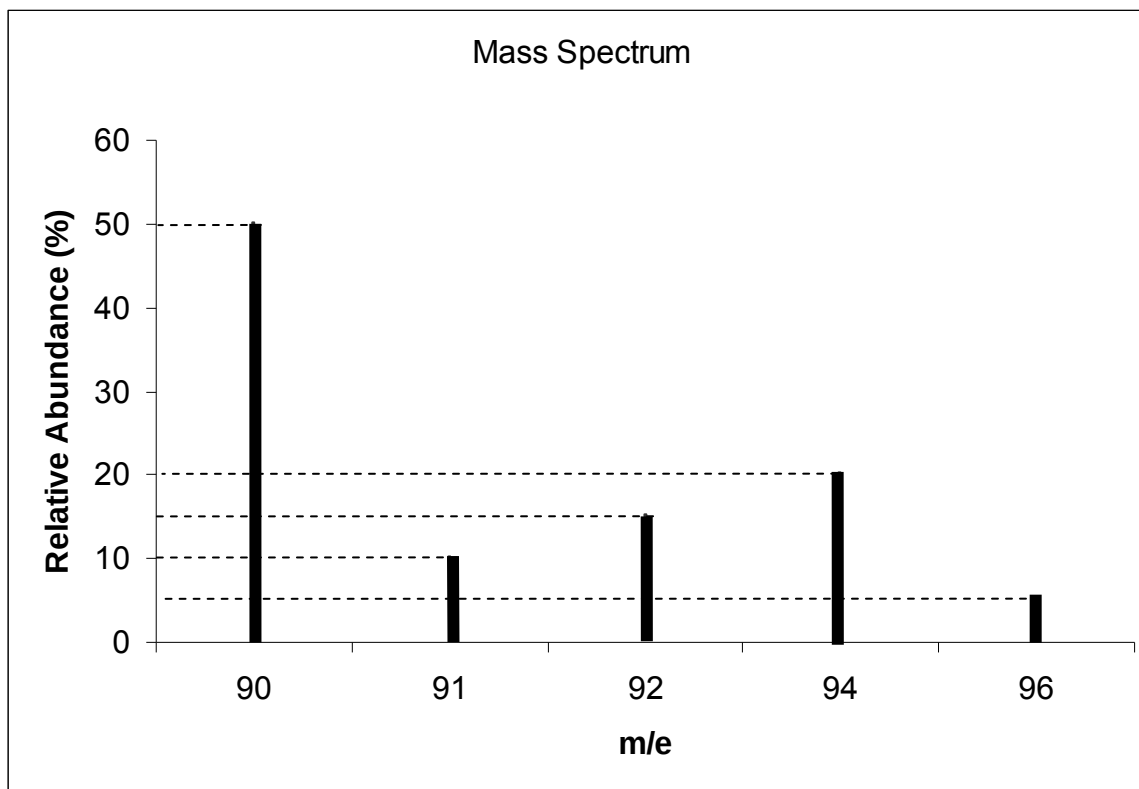
Date: 22/09/2011

This document consists of **17** printed pages.

Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 An element **A** consists of five isotopes. The mass spectrum of **A** is shown in the diagram.



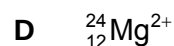
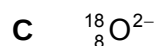
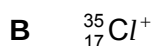
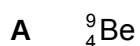
What is the relative atomic mass of **A**?

- A** 90.0
 - B** 91.5
 - C** 92.6
 - D** 93.0
- 2 A 25 cm³ sample of a 0.1 mol dm⁻³ **BCl**₂ solution requires 10 cm³ of 0.1 mol dm⁻³ of acidified potassium manganate(VII) solution for complete reaction.

What is the final oxidation state of element **B**?

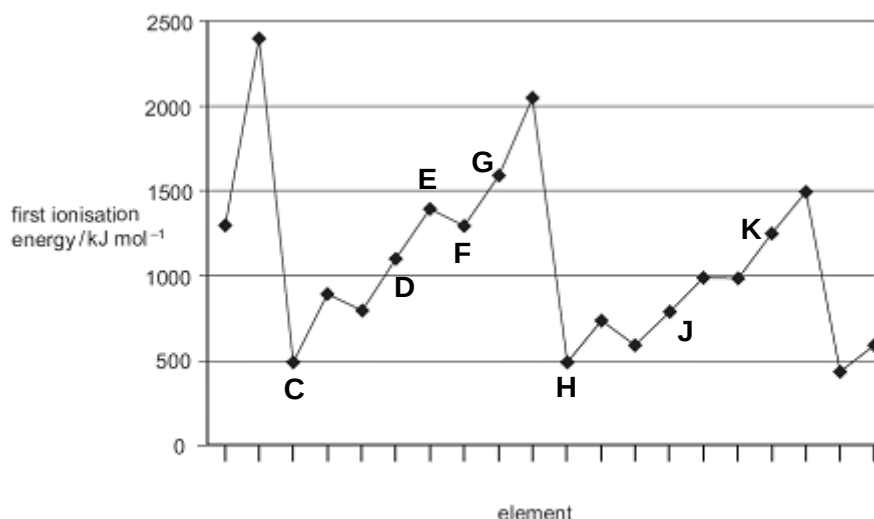
- A** +3
- B** +4
- C** +5
- D** +6

3 In which species are the number of electrons and neutrons equal?



4 Use of the Data Booklet is relevant to this question.

The first ionisation energies of successive elements in the Periodic Table are represented in the graph.



Which of the following statements about elements **C** to **K** is **incorrect**?

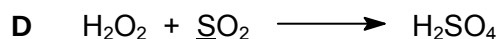
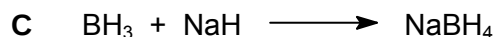
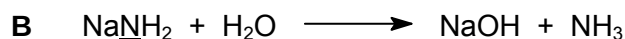
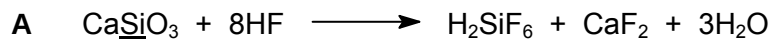
A **D** and **F** form compounds with the formulae **DF** and **DF₂**.

B **G** and **K** exhibit variable oxidation states from -1 to $+7$.

C **E** and **J** have half-filled p-orbitals.

D **H** has a larger atomic radius than **C**.

5 Which of the following reactions results in an increase in the bond angle of the underlined substance?



- 6 The co-ordination number of an ion is the number of oppositely charged ions that surround it in an ionic crystal lattice. This number depends mainly on two factors – the relative sizes of the ions and their relative charges. The table below shows the data for the ions in the lattices of NaCl, CsCl and MgO.

compound	cation radius / nm	anion radius / nm	$\frac{\text{anion radius}}{\text{cation radius}}$	co-ordination no. of cation	co-ordination no. of anion
NaCl	0.10	0.18	1.8	6	6
CsCl	0.17	0.18	1.1	8	8
MgO	0.07	0.14	2.0	6	6

Which of the following would be **unlikely** co-ordination numbers for the corresponding compound?

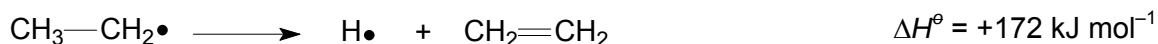
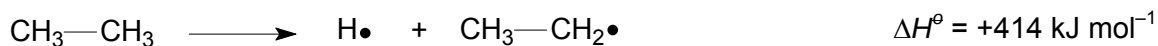
	compound	cation radius / nm	anion radius / nm	co-ordination no. of cation	co-ordination no. of anion
A	NaBr	0.10	0.20	6	6
B	BaO	0.14	0.14	8	8
C	CaF ₂	0.10	0.13	8	4
D	MgCl ₂	0.07	0.18	6	6

- 7 Two identical bulbs at the same temperature contain ideal gases **L** and **M** separately. The density of gas **L** is twice that of gas **M** and the molecular mass of gas **L** is half that of gas **M**.

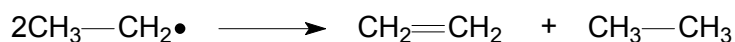
What is the ratio of the pressure of gas **L** to that of gas **M**?

- A 1 : 2 B 1 : 1 C 2 : 1 D 4 : 1

- 8 The standard enthalpy changes for two gas-phase reactions at constant pressure are shown below.



From these values, the standard enthalpy change, $\Delta H^\ominus$, for the reaction



can be calculated to be

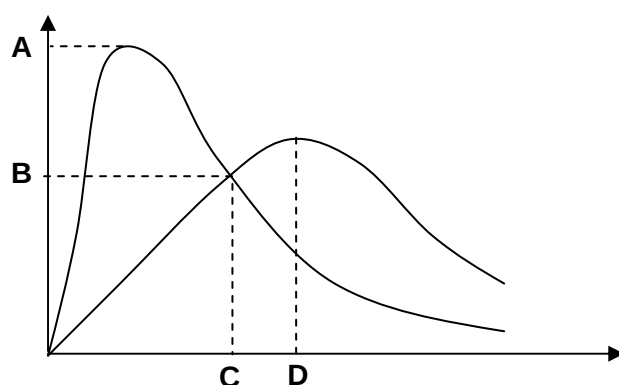
- A -242 kJ mol^{-1}
 B $+242 \text{ kJ mol}^{-1}$
 C -586 kJ mol^{-1}
 D $+586 \text{ kJ mol}^{-1}$

9 For which of the following reactions is the standard entropy change, ΔS° , positive?

- A $\text{H}_2\text{O}(\text{g}) \longrightarrow \text{H}_2\text{O}(\text{l})$
 B $\text{PCl}_3(\text{l}) + \text{Cl}_2(\text{g}) \longrightarrow \text{PCl}_5(\text{s})$
 C $2\text{Na}(\text{s}) + 2\text{H}_2\text{O}(\text{l}) \longrightarrow 2\text{NaOH}(\text{aq}) + \text{H}_2(\text{g})$
 D $\text{U}(\text{s}) + 3\text{F}_2(\text{g}) \longrightarrow \text{UF}_6(\text{s})$

10 Air inside a car engine can reach a temperature of about 500°C , enabling nitrogen to react with oxygen to form nitrogen oxides.

The diagram shows the Boltzmann distribution for air at room temperature and 500°C .



Which letter on the axes represents the most probable energy of the air molecules inside the car engine?

11 A molten chloride of a Group I metal, QCl , and a molten chloride of a Group II metal, RCl_2 , is separately electrolysed using the same current for the same period of time.

Which of the following statements about the experiment is correct?

- A An equal number of moles of **Q** and **R** are deposited.
 B The mass of elemental **Q** and **R** that are deposited are independent of the electrolysis time.
 C The number of moles of **R** deposited is half the number of moles of **Q** deposited.
 D RCl_2 gives off twice the volume of chlorine gas as QCl , measured under the same conditions of temperature and time.

12 In which of the following reactions will the position of equilibrium shift to the left when the pressure on the system is increased?

- A $\text{C}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons \text{CO}_2(\text{g})$
 B $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
 C $2\text{Mg}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{MgO}(\text{s})$
 D $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{H}_2\text{O}(\text{g})$

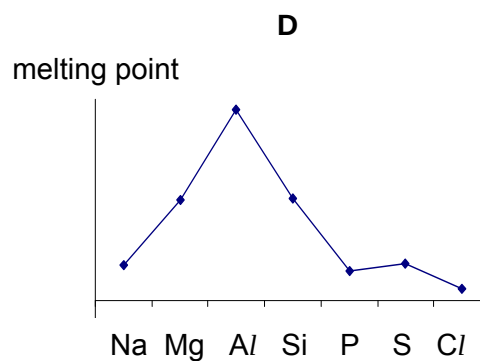
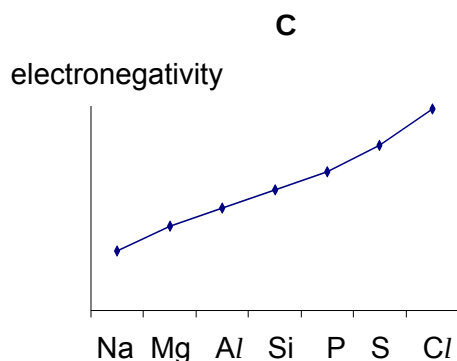
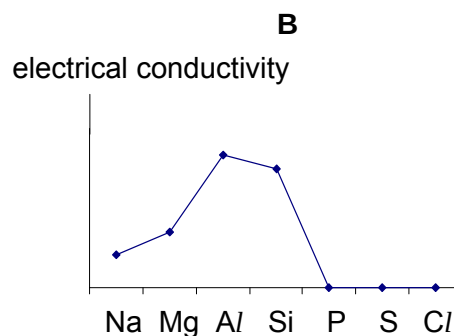
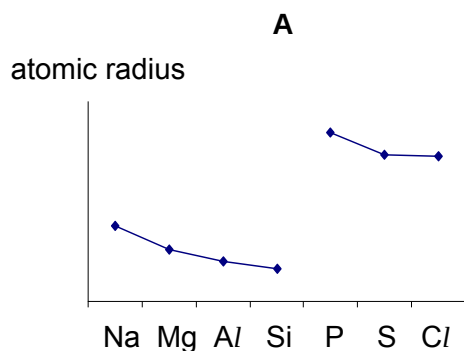
- 13 At body temperature of 37 °C, K_w has a value of 2.4×10^{-14} .

What is the concentration of OH^- if the pH of blood is 7.4 under these conditions?

- A 7.00×10^{-7}
 B 6.03×10^{-7}
 C 2.51×10^{-7}
 D 3.98×10^{-8}
- 14 A solution contains $0.03 \text{ mol dm}^{-3} \text{ Fe}^{2+}$ and $0.02 \text{ mol dm}^{-3} \text{ Ag}^+$ at 25 °C. The Fe^{2+} and Ag^+ can be separated by slowly adding Na_2CO_3 solid. The numerical value of the solubility product of FeCO_3 at 25 °C is 2.1×10^{-11} and that of Ag_2CO_3 is 8.1×10^{-12} .

At which point would the concentration of Fe^{2+} in the solution be half that of its original concentration?

- A after the first trace of FeCO_3 and before the first trace of Ag_2CO_3
 B after the first trace of Ag_2CO_3 and before the first trace of FeCO_3
 C at the first trace of FeCO_3
 D at the first trace of Ag_2CO_3
- 15 Which of the following sketches shows the correct trend in the stated property, for the elements in the third period of the Periodic Table?



- 16 A chloride and an oxide of the elements in the third period of the Periodic Table are dissolved in two separate portions of water.

Both the solutions prepared can be used to dissolve Al_2O_3 but only one can be used to dissolve SiO_2 .

Which of the following could be the chloride and the oxide used?

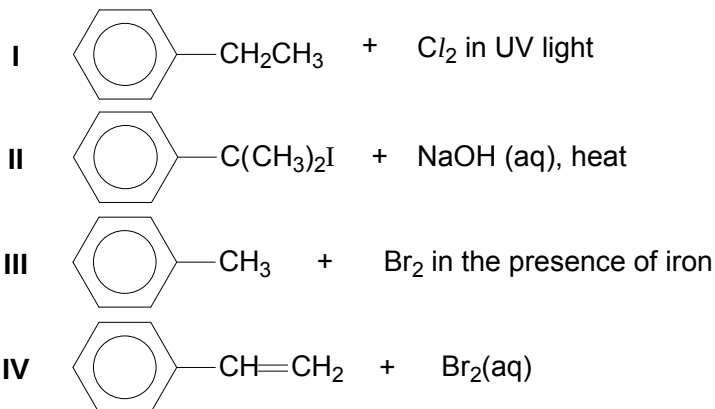
- | | | |
|----------|----------|-------------|
| A | PCl_5 | P_4O_{10} |
| B | $NaCl$ | MgO |
| C | $SiCl_4$ | Na_2O |
| D | $MgCl_2$ | SO_3 |
- 17 Which of the following statements concerning anhydrous magnesium nitrate and anhydrous barium nitrate is **incorrect**?
- A** Nitrogen dioxide is evolved at a lower temperature from magnesium nitrate than from barium nitrate when both are heated.
- B** The lattice energy of magnesium nitrate is more exothermic than that of barium nitrate.
- C** Upon heating both nitrates, the volume of nitrogen dioxide evolved is 4 times greater than the volume of oxygen.
- D** When both nitrates are added separately to water, solutions with the same pH are obtained.
- 18 Which of the following statements about the trends in the properties of the halogen down the group is correct?
- A** The electronegativity increases.
- B** The reactivity as oxidising agents increases.
- C** The enthalpy change of reaction with hydrogen becomes more endothermic.
- D** The volatility increases.
- 19 A green solid of general formula $Cr(H_2O)_6Cl_3$ dissolves readily in water to form an aqueous solution. When this solution is treated with an excess of aqueous silver nitrate, only two third of the total chloride present is precipitated as $AgCl$.

What represents the structure of the chromium ion present in the original compound?

- A** $[Cr(H_2O)_6]^{3+}$
- B** $[Cr(H_2O)_3Cl_3]$
- C** $[Cr(H_2O)_4Cl_2]^+$
- D** $[Cr(H_2O)_5Cl]^{2+}$

- 20 Consider the organic intermediate which will be formed in each of the reactions given below.

In which of the following pairs does the reactive carbon atom in the reactant change from being sp^3 hybridised to being sp^2 hybridised in the intermediate?

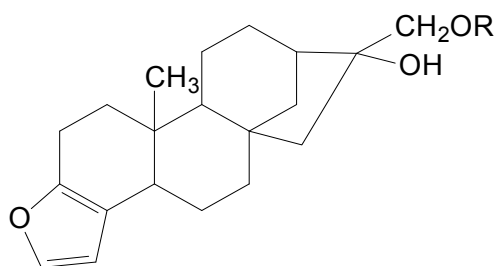


- A I and II
 B I and III
 C II and IV
 D III and IV
- 21 When alkane **T**, C_7H_{16} , was reacted with bromine under ultraviolet light, it produced only three isomeric monobromo compounds, with the formula $C_7H_{15}Br$.

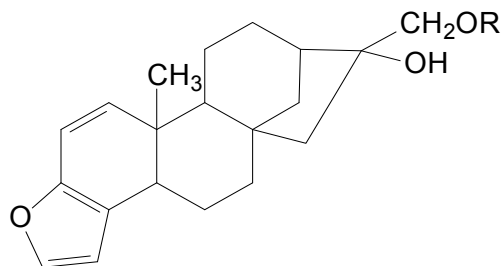
What is the likely identity of alkane **T**?

- A $CH_3(CH_2)_5CH_3$
 B $(CH_3)_2CH(CH_2)_3CH_3$
 C $(CH_3)_3CCH_2CH_2CH_3$
 D $(CH_3)_3CCH(CH_3)_2$

- 22 Cafestol and Kahweol are fat-soluble compounds known as diterpenes, which are present in the oil derived from coffee beans. Both compounds have liver-protecting properties of caffeine. The structures of Cafestol and Kahweol are given below.



Cafestol

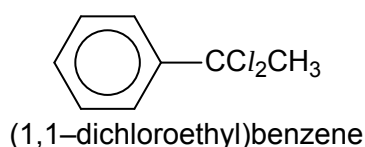


Kahweol

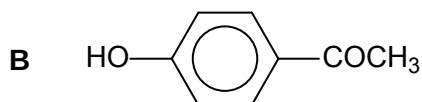
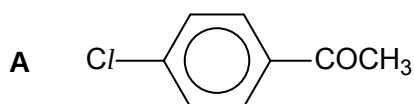
In the following reactions, $-\text{CH}_2\text{OR}$ remains unaltered.

Which of the following statements is true?

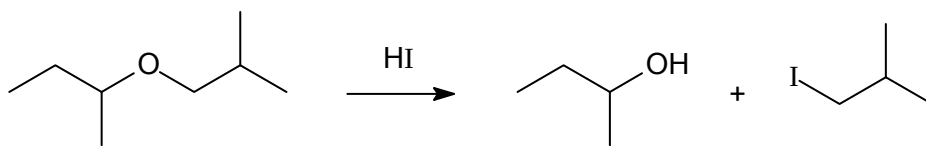
- A** Both compounds do not react with acidified $\text{K}_2\text{Cr}_2\text{O}_7$.
- B** 1 mol of Cafestol reacts with excess Na to produce 1 mol of H_2 .
- C** 1 mol of Kahweol reacts with Br_2 to give a product with 11 chiral centres.
- D** When treated with an excess of hot concentrated acidified KMnO_4 , Cafestol will produce a compound with 4 carbonyl groups.
- 23 Chlorine, in the presence of aluminium chloride, undergoes a substitution reaction with benzene forming chlorobenzene. When (1,1-dichloroethyl)benzene reacts with iodine monochloride, ICl , in the presence of aluminium chloride, a similar reaction occurs and compound **U** is formed.



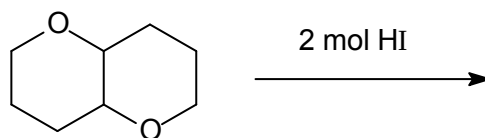
Which product will be present in the greatest yield when **U** is heated under reflux with aqueous sodium hydroxide?



24 Ethers are cleaved by HI via an S_N2 mechanism.

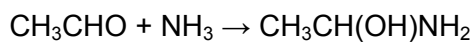


Which of the following is the most likely product for the following reaction?



- A**
- B**
- C**
- D**

25 Ethanal can react with ammonia as shown.



What type of reaction is this?

- A** addition–elimination
- B** electrophilic addition
- C** free–radical addition
- D** nucleophilic addition

- 26 Compounds **V** and **W** both give an orange precipitate with 2,4–dinitrophenylhydrazine. Only one of the compounds gives a yellow precipitate with alkaline aqueous iodine and also decolourises hot acidified potassium manganate(VII).

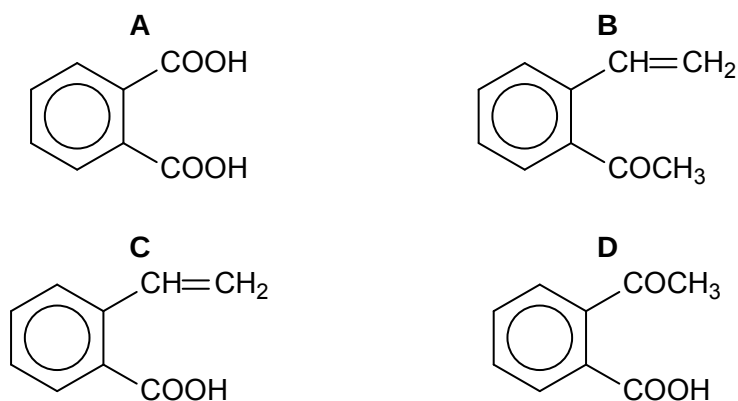
Which of the following pairs could **V** and **W** be?

	V	W
A	CH_3COCH_3	$\text{CH}_3\text{CH}_2\text{CHO}$
B	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	$\text{CH}_3\text{CH}_2\text{CHO}$
C	$\text{CH}_3\text{CH}(\text{OH})\text{COCH}_3$	$\text{CH}_2=\text{CHCOCH}_3$
D	$\text{CH}_3(\text{CH}_2)_2\text{CHO}$	$\text{CH}_2=\text{CHCOCH}_3$

- 27 An organic compound **X** has the following properties:

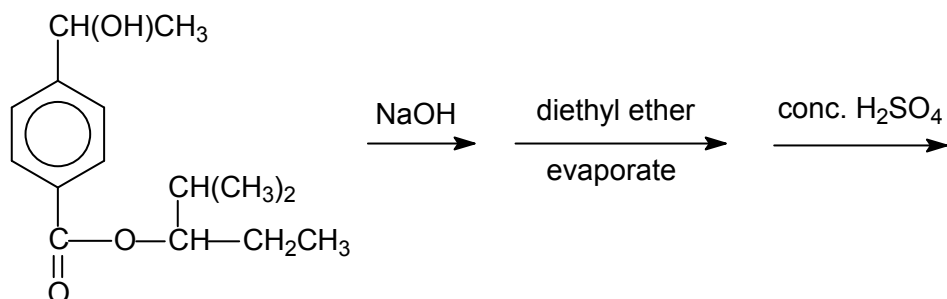
- It has molecular formula $\text{C}_9\text{H}_{10}\text{O}_2$.
- It gives a silver mirror with Tollens' reagent.
- It gives yellow crystals with alkaline aqueous iodine.
- Heating **X** with Al_2O_3 at 300°C , produces a compound **Y** which gives an effervescence with hot acidified KMnO_4 .

Which of the following is formed when **Y** is oxidised by acidified $\text{K}_2\text{Cr}_2\text{O}_7$?



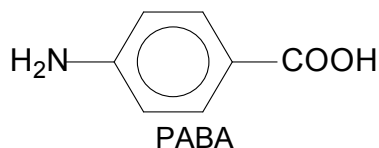
- 28 The following ester was hydrolysed by boiling with aqueous sodium hydroxide, and the resulting organic products were shaken with diethyl ether (an organic solvent). After allowing the mixture to stand, the organic layer was separated. The diethyl ether was then evaporated to leave a residue.

This residue was subjected to a dehydration reaction using concentrated H_2SO_4 and a final product mixture was obtained.



What is the number of isomers present in the final mixture?

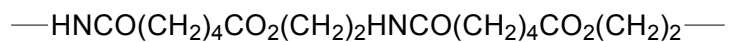
- A 1 B 2 C 3 D 4
- 29 4-aminobenzoic acid, commonly known as PABA, is used as a drug against fibrotic skin disorders, and can also be used in the treatment of irritable bowel syndrome.



Which of the following is a possible means of its preparation?

- A $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{Sn, conc HCl}}$ $\xrightarrow[\text{heat}]{\text{H}_2\text{O} / \text{H}^+}$
- B $\xrightarrow[\text{heat}]{\text{H}_2\text{O} / \text{H}^+}$ $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{Sn, conc HCl}}$
- C $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{Sn, conc HCl}}$ $\xrightarrow[\text{heat}]{\text{KMnO}_4 / \text{H}^+}$
- D $\xrightarrow[\text{heat}]{\text{KMnO}_4 / \text{H}^+}$ $\xrightarrow[\text{conc H}_2\text{SO}_4]{\text{conc HNO}_3}$ $\xrightarrow{\text{Sn, conc HCl}}$

30 Part of the structure of a polymer is shown below.



Which of the following statements about the polymer is correct?

- A It can be made from $\text{ClCO(CH}_2\text{)}_2\text{COCl}$ and $\text{HOCH}_2\text{(CH}_2\text{)}_2\text{CH}_2\text{NH}_2$.
- B It is both a polyester and a polyamide.
- C It reacts with HCN in a trace of base.
- D It is resistant to alkaline hydrolysis.

Section B

For each of the question in this section, one or more of the three numbered statements 1 to 3 may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

- 31** A monomer undergoes addition polymerisation. A 1 mol sample of the monomer is completely polymerised.

How many moles of polymer might, theoretically, be formed?

- 1** 1
2 10^{-6}
3 $\frac{1}{6.02 \times 10^{23}}$

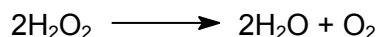
- 32** Which of the following property is a result of intermolecular hydrogen bonding?

- 1** Ice has a lower density than water at 0 °C.
2 The bond angle about the oxygen atom in ice is 109.5°.
3 The relative molecular mass of ethanoic acid, when dissolved in benzene, is 120.

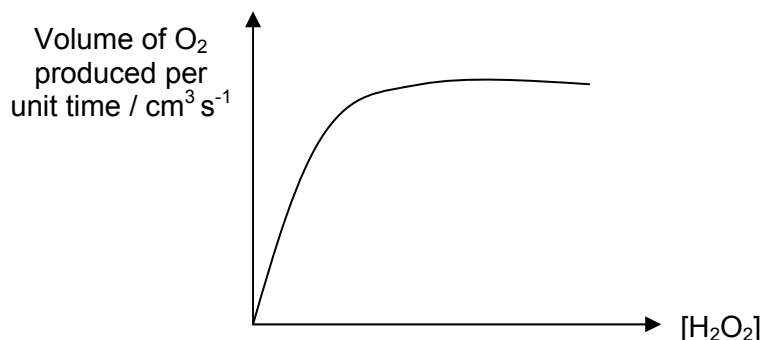
- 33** Which of the following processes is always exothermic?

- 1** condensation of steam
2 diluting a solution with water
3 electrolysis of water

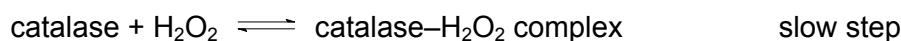
- 34 The rate of decomposition of aqueous H_2O_2 increases in the presence of the catalase enzyme.



The following graph was obtained in a study to investigate the effect of catalase on the rate of decomposition of H_2O_2 .



The steps predicted for this enzymatic decomposition of H_2O_2 are shown below.



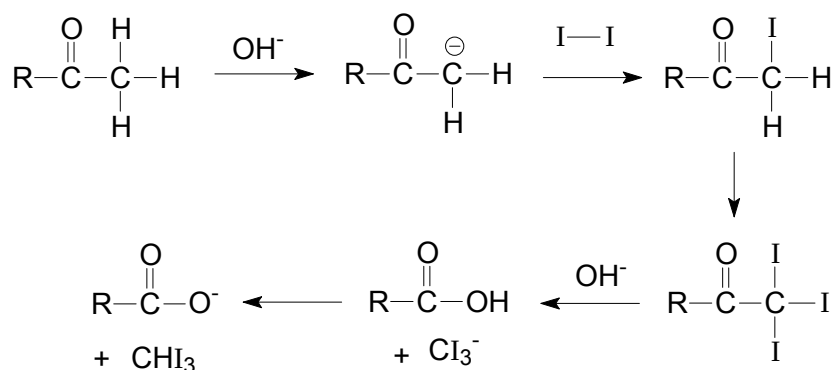
Which of the following statements about the enzymatic decomposition of H_2O_2 are correct?

- 1 At low $[\text{H}_2\text{O}_2]$, the rate of decomposition increases with increasing amount of catalase used.
 - 2 At low $[\text{H}_2\text{O}_2]$, catalase lowers the activation energy of the decomposition by forming temporary bonds with H_2O_2 molecules.
 - 3 At high $[\text{H}_2\text{O}_2]$, all the active sites of the catalase enzyme are saturated with H_2O_2 molecules.
- 35 Which statements are true about the elements in Group II of the Periodic Table?
- 1 They are reducing agent.
 - 2 Their ionic radii increase down the Group.
 - 3 Their electronegativities decrease down the Group.
- 36 *Use of the Data Booklet is relevant to this question.*

Which of the following statements are correct for all three halogens, chlorine, bromine and iodine?

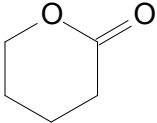
- 1 They all form hydrides that are strong acids in aqueous solutions.
- 2 They all react with aqueous sodium hydroxide to form halides and oxo-anions.
- 3 They all react with aqueous iron(II) sulfate to form iron(III) ions and halides.

37 The mechanism of the iodoform test is given below.



What are the types of reactions shown in this mechanism?

- 1 nucleophilic substitution
 - 2 acid-base reaction
 - 3 elimination
- 38 Which of the following statements are true in the comparison of 2-bromoethanoic acid and 2-chloroethanoic acid?
- 1 2-chloroethanoic acid is a stronger acid than 2-bromoethanoic acid.
 - 2 Equimolar amounts of both acids will react with ammonia to form the same amount of salt.
 - 3 Precipitation occurs at a faster rate for 2-bromoethanoic acid when both acids are reacted with ethanolic AgNO_3 .
- 39 Which of the following synthesis would yield the product shown?

- 1 $\text{HO}(\text{CH}_2)_4\text{COOH} \xrightarrow[\text{heat}]{\text{conc. H}_2\text{SO}_4}$ 
- 2 $\text{CH}_3\text{CH}_2\text{OH} \xrightarrow[\text{heat}]{\text{NaBr}} \text{CH}_3\text{CH}_2\text{Br}$
- 3 $\text{C}_6\text{H}_5\text{C}(\text{CH}_3)_3 \xrightarrow[\text{heat}]{\text{KMnO}_4, \text{H}^+} \text{C}_6\text{H}_5\text{COOH}$

40 The pK_a values of some ammonium ions at 298 K are given below.

Ammonium ion	pK_a
CH_3NH_3^+	10.66
$(\text{CH}_3)_3\text{NH}^+$	9.81

Which of the following statements are correct?

- 1 The pH of 0.1 mol dm^{-3} of $(\text{CH}_3)_3\text{N}$ is 11.4.
- 2 $(\text{CH}_3)_3\text{N}$ is more basic than CH_3NH_2 due to the presence of more electron donating alkyl groups.
- 3 1 mol of $(\text{CH}_3)_3\text{N}$ requires more HCl for complete neutralization than 1 mol of CH_3NH_2 .

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	D
2	B	22	A
3	C	23	C
4	B	24	B
5	B	25	D
6	D	26	D
7	D	27	C
8	A	28	C
9	C	29	C
10	D	30	B
11	C	31	C
12	B	32	A
13	B	33	D
14	A	34	A
15	C	35	A
16	C	36	B
17	D	37	B
18	C	38	A
19	D	39	D
20	A	40	D